2.4.6

EE25BTECH11016 - Taraka Abhinav

Question:
If $\mathbf{a} = 5\hat{i} - \hat{j} - 3\hat{k}$ and $\mathbf{b} = \hat{i} + 3\hat{j} - 5\hat{k}$, then show that the vectors $\mathbf{a} + \mathbf{b}$ and $\mathbf{a} - \mathbf{b}$ are perpendicular.

Solution:

Two vectors are perpendicular if their dot product is zero. We need to show that $(\mathbf{a} + \mathbf{b})^T(\mathbf{a} - \mathbf{b}) = 0$.

First, we expand the algebraic expression for the dot product:

$$\begin{aligned}(\mathbf{a} + \mathbf{b})^T(\mathbf{a} - \mathbf{b}) &= (\mathbf{a}^T + \mathbf{b}^T)(\mathbf{a} - \mathbf{b}) \\ &= \mathbf{a}^T\mathbf{a} - \mathbf{a}^T\mathbf{b} + \mathbf{b}^T\mathbf{a} - \mathbf{b}^T\mathbf{b} \end{aligned} \tag{1}$$

Since the dot product is commutative, $\mathbf{a}^T\mathbf{b} = \mathbf{b}^T\mathbf{a}$. The expression simplifies to:

$$(\mathbf{a} + \mathbf{b})^T(\mathbf{a} - \mathbf{b}) = \mathbf{a}^T\mathbf{a} - \mathbf{b}^T\mathbf{b} = |\mathbf{a}|^2 - |\mathbf{b}|^2 \tag{2}$$

Now, we calculate the squared magnitudes of $\mathbf{a}$ and $\mathbf{b}$ using their numerical values. The given vectors are:

$$\mathbf{a} = \begin{pmatrix} 5 \\ -1 \\ -3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 3 \\ -5 \end{pmatrix} \tag{3}$$

The squared magnitudes are:

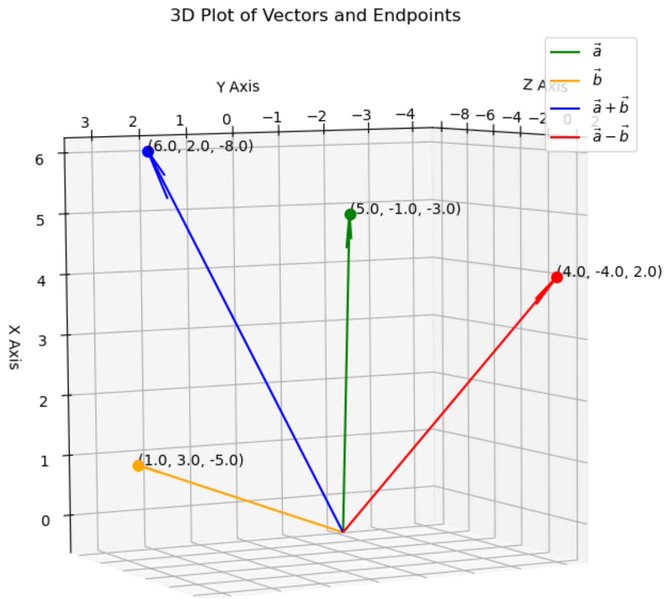
$$\begin{aligned}|\mathbf{a}|^2 = \mathbf{a}^T\mathbf{a} &= (5)^2 + (-1)^2 + (-3)^2 \\ &= 25 + 1 + 9 = 35 \end{aligned} \tag{4}$$

$$\begin{aligned}|\mathbf{b}|^2 = \mathbf{b}^T\mathbf{b} &= (1)^2 + (3)^2 + (-5)^2 \\ &= 1 + 9 + 25 = 35 \end{aligned} \tag{5}$$

Substituting the results from (4) and (5) into our algebraic result from (2):

$$(\mathbf{a} + \mathbf{b})^T(\mathbf{a} - \mathbf{b}) = 35 - 35 = 0 \tag{6}$$

Since the dot product is zero, the vectors $(\mathbf{a} + \mathbf{b})$ and $(\mathbf{a} - \mathbf{b})$ are perpendicular. Hence, proved.


3D Plot of Vectors and Endpoints
Y Axis
Z Axis
X Axis
(6.0, 2.0, -8.0)
(5.0, -1.0, -3.0)
(4.0, -4.0, 2.0)
(1.0, 3.0, -5.0)
$\vec{a}$
$\vec{b}$
$\vec{a}+\vec{b}$
$\vec{a}-\vec{b}$